

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Third Semester of B. Tech (EC) Examination****December 2016****EC 201.01 / EC 201 Digital Electronics & Logic Design****Date: 14.12.2016, Wednesday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the below questions. [07]**

- a. Which of the following circuit is sequential logic type design? [01]
 (a) Multiplexer (c) Full Adder
 (b) Johnson Counter (d) Magnitude Comparator
- b. Which of the following is MSI type IC? [01]
 (a) 74LS32 (c) 74LS86
 (b) 74LS85 (d) 74LS00
- c. Which gates are referring as Universal Gates? Also justify answer. [02]
- d. Simplify the Boolean function using K-map and draw logic diagram for simplified Function. $F(x,y,z) = \sum(0,2,3,4,5,6)$ [03]

Q – 2 Answer the below questions. [14]

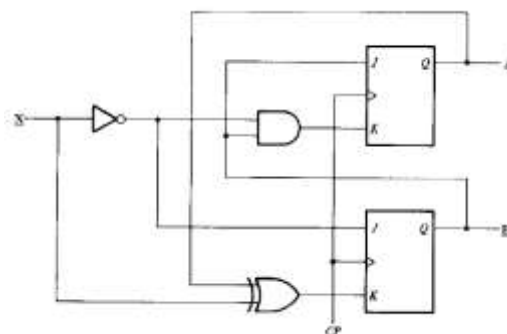
- a. Reduce following function and implement reduced function using NAND Gates only. [07]
 $F(A,B,C,D) = \sum(1,3,7,11,15) + d(0,2,5)$
- b. Draw and explain Logic circuit of Bidirectional Shift Register with Parallel load. [07]

OR**Q – 2 Answer the below questions. [14]**

- a. Simplify the following function using tabulation method and implement reduced function using NAND Gates only. $F(A,B,C,D) = \sum(0,1,2,8,10,11,14,15)$ [07]
- b. Draw and explain Logic circuit of BCD Ripple counter. [07]

Q – 3 Answer the below questions. [14]

- a. Write state table and state diagram for following sequential circuit. [07]



- b. Explain 4-bit Magnitude comparator with designing equations and draw 2-bit [07]

magnitude comparator combinational logic circuit.

OR

- Q – 3 Answer the below questions.** [14]
a. Design a 3-bit counter that has a repeated sequence of six states as 000, 001, 010, 100, 101 and 110. [07]
b. Draw and explain block diagram of a BCD adder circuit with require truth table. [07]

SECTION – II

- Q - 4 Answer the below questions.** [07]
a. The number of flip-flops required to build an MOD-19 counter is [01]
 (a) 4 (c) 6
 (b) 5 (d) 7
b. Which of the following statements is not correct? [01]
 (a) ECL has the least propagation delay (c) I²L logic is built-in function
 (b) TTL has the largest fan-out (d) CMOS has the largest noise-region
c. Implement half adder combinational logic diagram using NOR Gates only. [02]
d. Write the excitation table for RS, JK and T flip-flops. [03]
- Q – 5 Answer the below questions.** [14]
a. Write a short note on switch-tail ring counter. [07]
b. Design BCD to Excess 3 Code Converter. [07]

OR

- Q – 5 Answer the below questions.** [14]
a. Write a short note on memory unit with its functioning for read and write function. [07]
b. Design Full Sub-tractor using with truth table and combinational logic circuit. [07]
- Q – 6 Answer the below questions.** [14]
a. Explain operation of D-type positive edge triggered flip-flop with logic circuit. [07]
b. Implement following function using proper multiplexer. Use A, B and D as select line of multiplexer. $F(A,B,C,D) = \sum(0,1,2,8,10,11,14,15)$ [07]

OR

- Q – 6 Answer the below questions.** [14]
a. Explain Carry Propagation and draw logic diagram of a look ahead carry generator. [07]
b. Reduce following state diagram and rewrite reduced state table and state diagram. [07]

